

Question 1

Prove that there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers (repetitions are allowed).

For the sake of contradiction, assume there exists a positive perfect cube n^3 less than 1000 that is the sum of the cubes of two positive integers a and b such that $a^3 + b^3 = n^3$.

since $n^3 < 1000$, $n < 10$ which implies n is an element of the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Let's consider each possible value of n :

1. For $n = 1$, there are no positive integers a and b such that $a^3 + b^3 = 1^3 = 1$.
2. For $n = 2$, there are no positive integers a and b such that $a^3 + b^3 = 2^3 = 8$.
3. For $n = 3$, there are no positive integers a and b such that $a^3 + b^3 = 3^3 = 27$.
4. For $n = 4$, there are no positive integers a and b such that $a^3 + b^3 = 4^3 = 64$.
5. For $n = 5$, there are no positive integers a and b such that $a^3 + b^3 = 5^3 = 125$.
6. For $n = 6$, there are no positive integers a and b such that $a^3 + b^3 = 6^3 = 216$.
7. For $n = 7$, there are no positive integers a and b such that $a^3 + b^3 = 7^3 = 343$.
8. For $n = 8$, there are no positive integers a and b such that $a^3 + b^3 = 8^3 = 512$.
9. For $n = 9$, there are no positive integers a and b such that $a^3 + b^3 = 9^3 = 729$.

Therefore, there are no positive perfect cubes less than 1000 that are the sum of the cubes of two positive integers. ■

Question 2

Use a proof by cases to show that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a, b , and c are real numbers.

To prove that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a, b , and c are real numbers, we will consider the following cases:

Case 1: $a \leq b$ and $a \leq c$ If a is the smallest of the three numbers, then $\min(a, b) = a$ and $\min(a, c) = a$.

Therefore, $\min(\min(a, b), c) = \min(a, c) = a$. Similarly, $\min(a, \min(b, c)) = \min(a, b) = a$.

Therefore, $\min(a, \min(b, c)) = \min(\min(a, b), c)$ when $a \leq b$ and $a \leq c$.

Case 2: $b \leq a$ and $b \leq c$ If b is the smallest of the three numbers, then $\min(a, b) = b$ and $\min(b, c) = b$.

Therefore, $\min(\min(a, b), c) = \min(b, c) = b$. Similarly, $\min(a, \min(b, c)) = \min(a, b) = b$.

Therefore, $\min(a, \min(b, c)) = \min(\min(a, b), c)$ when $b \leq a$ and $b \leq c$.

Case 3: $c \leq a$ and $c \leq b$ If c is the smallest of the three numbers, then $\min(a, c) = c$ and $\min(b, c) = c$.

Therefore, $\min(\min(a, b), c) = \min(a, c) = c$. Similarly, $\min(a, \min(b, c)) = \min(a, c) = c$.

Therefore, $\min(a, \min(b, c)) = \min(\min(a, b), c)$ when $c \leq a$ and $c \leq b$.

Since these cases cover all possibilities, we have shown that $\min(a, \min(b, c)) = \min(\min(a, b), c)$ whenever a, b , and c are real numbers.

■

Question 3

Prove using the notion without loss of generality that $5x + 5y$ is an odd integer when x and y are integers of opposite parity (one is even and another is odd).

Without loss of generality, assume that x is even and y is odd.

Then, by definition, there exist integers m and n such that $x = 2m$ and $y = 2n + 1$.

Substitute $x = 2m$ and $y = 2n + 1$ into the equation $5x + 5y$:

$$5(2m) + 5(2n + 1) = 10m + 10n + 5 = 2(5m + 5n + 2) + 1$$

Since integers are closed under addition and multiplication, $5m + 5n + 2$ is an integer and $2(5m + 5n + 2)$ is even. Thus, $2(5m + 5n + 2) + 1$ is odd.

Therefore, $5x + 5y$ is an odd integer when x and y are integers of opposite parity. ■

Question 4

Prove that there is a positive integer x that equals the sum of the positive integers less than x . Is your proof constructive or nonconstructive?

Consider the integer 3.

The series of positive integers less than 3 is $1 + 2 = 3$.

Therefore, 3 equals the sum of the positive integers less than 3.

Therefore, there exists a positive integer x that equals the sum of the positive integers less than x . ■

This proof is constructive, as it provides a specific example ($x = 3$) that satisfies the condition.

Question 5

Prove that either $2 * 10^{500} + 15$ or $2 * 10^{500} + 16$ is not a perfect square. Is your proof constructive or nonconstructive?

For any two consecutive integers n and $n + 1$, the difference between the squares of these two integers is $2n + 1$.

Thus, two consecutive perfect squares can be expressed as n^2 and $(n + 1)^2$.

The difference between two consecutive perfect squares is then found by finding the difference between these two expressions:

$$(n + 1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$$

When $n \geq 0$, the difference between the squares of two consecutive integers is always greater than 1.

Now assume for the sake of contradiction that both $2 * 10^{500} + 15$ and $2 * 10^{500} + 16$ are perfect squares.

Since $\sqrt{2 * 10^{500} + 15}$ and $\sqrt{2 * 10^{500} + 16}$ are both greater than 0, the difference between them must be greater than 1.

However, the difference between $2 * 10^{500} + 15$ and $2 * 10^{500} + 16$ is 1.

Therefore, either $2 * 10^{500} + 15$ or $2 * 10^{500} + 16$ is not a perfect square. ■

This proof does not explicitly construct a specific integer or provide an example of a non-perfect square among the two numbers by way of direct evaluation. Instead, it relies on the contradiction between the properties of perfect squares and the specific difference observed. Therefore, it qualifies as a nonconstructive proof.

Question 6

Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or nonconstructive? [Hint: Do not try to evaluate these numbers!]

Find the sign of the first factor: $65^{1000} > 8^{2001} \implies 65^{1000} - 8^{2001} + 3^{177} > 0$

Find the sign of the second factor: $79^{1212} > 9^{2399} \implies 79^{1212} - 9^{2399} + 2^{2001} > 0$

Find the sign of the third factor: $24^{4493} > 5^{8192} \implies 24^{4493} - 5^{8192} + 7^{1777} > 0$

The product of two nonnegative numbers is nonnegative. Since all three factors are nonnegative, the product of any two of these numbers is nonnegative. ■

This proof is nonconstructive, as it does not explicitly construct a specific example or provide an example of a nonnegative product by way of direct evaluation. Instead, it relies on the properties of the numbers and their signs to determine the nonnegativity of the product.

Question 7

Prove or disprove that if a and b are rational numbers, then ab is also rational.

By the *Closure property of rational numbers under multiplication*, which is an axiom, it is understood that rational numbers are closed under multiplication. Therefore, if a and b are rational numbers, then ab is also rational. Hence, ab being rational does not require proof.

If ab being rational required proof, it would suggest that the closure property of rational numbers under multiplication is derived from other properties, disqualifying it as an axiom. However, the closure property of rational numbers under multiplication is an axiom and does not require proof.

Therefore, if a and b are rational numbers, then ab is also rational. ■

Question 8

Show that if a , b , and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax + b = c$.

Given the equation $ax + b = c$, we can solve for x :

$$ax = c - b$$

$$x = \frac{c-b}{a}$$

Since $a \neq 0$, the denominator is nonzero, and the solution for x is unique.

If it were the case that there was another possible solution for x , then:

$$\frac{c-b}{a} = \frac{c-b}{a}$$

But since these are equal, there is only one unique solution for x .

Therefore, if a , b , and c are real numbers and $a \neq 0$, then there is a unique solution of the equation $ax + b = c$. ■

Question 9

Show that if r is an irrational number, there is a unique integer n such that the distance between r and n is less than $\frac{1}{2}$.

Since r exists on a number line, it will always be equal to an integer n plus a fraction f such that $0 \leq f < 1$.

When f is less than $\frac{1}{2}$, the distance between r and n is less than $\frac{1}{2}$.

When f is greater than or equal to $\frac{1}{2}$, the distance between r and n is greater than $\frac{1}{2}$.

If f is exactly $\frac{1}{2}$, then the number could be written as $\frac{n}{2} + \frac{1}{2} = \frac{n+1}{2}$. Since integers are closed under multiplication and addition, $n+1$ is an integer, and $\frac{n+1}{2}$ is therefore a ratio of two integers, making r rational. Thus, f cannot be exactly $\frac{1}{2}$.

To prove uniqueness, assume for the sake of contradiction that there exist two integers n_1 and n_2 such that the distance between r and n_1 is less than $\frac{1}{2}$ and the distance between r and n_2 is less than $\frac{1}{2}$.

Then, $|r - n_1| < \frac{1}{2}$ and $|r - n_2| < \frac{1}{2}$.

This implies that n_1 and n_2 are within $\frac{1}{2}$ of r , which means they are within $\frac{1}{2}$ of each other.

However, the distance between two integers is always an integer, and the distance between n_1 and n_2 is less than $\frac{1}{2}$, which is not possible.

Therefore, there cannot be two integers n_1 and n_2 such that the distance between r and n_1 is less than $\frac{1}{2}$ and the distance between r and n_2 is less than $\frac{1}{2}$.

Therefore, there is a unique integer n such that the distance between r and n is less than $\frac{1}{2}$. ■

Question 10

Prove that given a real number x there exist unique numbers n and ε such that $x = n + \varepsilon$, n is an integer, and $0 \leq \varepsilon < 1$.

Given a real number x , let $n = \lfloor x \rfloor$ be the greatest integer less than or equal to x .

Let $\varepsilon = x - n$.

By construction, $n \leq x < n + 1 \implies 0 \leq x - n < 1 \implies 0 \leq \varepsilon < 1$.

Therefore, $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$.

To show uniqueness, assume that there exist integers n_1 and n_2 and real numbers ε_1 and ε_2 such that $x = n_1 + \varepsilon_1 = n_2 + \varepsilon_2$, where $n_1 \neq n_2$ or $\varepsilon_1 \neq \varepsilon_2$.

Without loss of generality, assume $n_1 < n_2$.

Then, $n_1 + \varepsilon_1 < n_2 + \varepsilon_2 \implies \varepsilon_1 < \varepsilon_2$.

Since ε_1 and ε_2 are real numbers between 0 and 1, $\varepsilon_2 - \varepsilon_1$ is a real number between 0 and 1.

However, $\varepsilon_2 - \varepsilon_1 = (n_1 - n_2) + (\varepsilon_1 - \varepsilon_2)$, which is an integer plus a real number between -1 and 1 .

This contradicts the fact that $\varepsilon_2 - \varepsilon_1$ is a real number between 0 and 1.

Therefore, the numbers n and ε are unique. ■

Question 11

Use forward reasoning to show that if x is a nonzero real number, then $x^2 + \frac{1}{x^2} \geq 2$. [Hint: Start with the inequality $(x - \frac{1}{x})^2 \geq 0$ which holds for all nonzero real numbers x .]

Given the inequality $(x - \frac{1}{x})^2 \geq 0$, expand the left side:

$$(x - \frac{1}{x})^2 = x^2 - 2 + \frac{1}{x^2} \geq 0$$

Add 2 to both sides:

$$x^2 + \frac{1}{x^2} \geq 2$$

Therefore, if x is a nonzero real number, then $x^2 + \frac{1}{x^2} \geq 2$. ■

Question 12

The harmonic mean of two real numbers x and y equals $\frac{2xy}{x+y}$. By computing the harmonic and geometric means of different pairs of positive real numbers, formulate a conjecture about their relative sizes and prove your conjecture.

Let x and y be positive real numbers. The harmonic mean of x and y is $\frac{2xy}{x+y}$. The geometric mean of x and y is $\sqrt{xy}$.

Testing Pairs

1. Let $x = 2$ and $y = 8$. Harmonic mean: $\frac{2 \cdot 8}{2+8} = \frac{16}{10} = 1.6$
Geometric mean: $\sqrt{2 \cdot 8} = \sqrt{16} = 4$
2. Let $x = 3$ and $y = 9$. Harmonic mean: $\frac{2 \cdot 3 \cdot 9}{3+9} = \frac{54}{12} = 4.5$.
Geometric mean: $\sqrt{3 \cdot 9} = \sqrt{27} \approx 5.196$
3. Let $x = 4$ and $y = 16$. Harmonic mean: $\frac{2 \cdot 4 \cdot 16}{4+16} = \frac{128}{20} = 6.4$
Geometric mean: $\sqrt{4 \cdot 16} = \sqrt{64} = 8$

Conjecture The geometric mean of two positive real numbers is always greater than or equal to the harmonic mean of the same two numbers.

Proof Given two positive real numbers x and y , the geometric mean is $\sqrt{xy}$ and the harmonic mean is $\frac{2xy}{x+y}$.

To prove that the geometric mean is greater than or equal to the harmonic mean, we need to show that $\sqrt{xy} \geq \frac{2xy}{x+y}$.

Squaring both sides, we get $xy \geq \frac{4x^2y^2}{(x+y)^2}$. Multiplying both sides by $(x+y)^2$, we get $xy(x+y)^2 \geq 4x^2y^2$. Expanding the left side, we get $x^3y + 2x^2y^2 + xy^3 \geq 4x^2y^2$. Rearranging terms, we get $x^3y + xy^3 \geq 2x^2y^2$. Dividing by xy , we get $x^2 + y^2 \geq 2xy$.

This is true by the AM-GM inequality, which states that the arithmetic mean of two nonnegative numbers is always greater than or equal to the geometric mean of the same two numbers.

Therefore, the geometric mean of two positive real numbers is always greater than or equal to the harmonic mean of the same two numbers.

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Question 13

Formulate a conjecture about the decimal digits that appear as the final decimal digit of the fourth power of an integer. Prove your conjecture using a proof by case.

Conjecture The final decimal digit of the fourth power of an integer can only be 0, 1, 5, or 6.

Proof To prove this conjecture, we will consider the possible final decimal digits of the fourth power of an integer in each case:

1. The final decimal digit of the integer is 0. Then the final decimal digit of the fourth power of the integer will also be 0.
2. The final decimal digit of the integer is 1. Then the final decimal digit of the fourth power of the integer will also be 1.
3. The final decimal digit of the integer is 2. Then the final decimal digit of the fourth power of the integer will be 6.
4. The final decimal digit of the integer is 3. Then the final decimal digit of the fourth power of the integer will be 1.
5. The final decimal digit of the integer is 4. Then the final decimal digit of the fourth power of the integer will be 6.
6. The final decimal digit of the integer is 5. Then the final decimal digit of the fourth power of the integer will also be 5.
7. The final decimal digit of the integer is 6. Then the final decimal digit of the fourth power of the integer will also be 6.
8. The final decimal digit of the integer is 7. Then the final decimal digit of the fourth power of the integer will be 1.
9. The final decimal digit of the integer is 8. Then the final decimal digit of the fourth power of the integer will be 6.
10. The final decimal digit of the integer is 9. Then the final decimal digit of the fourth power of the integer will be 1.

Therefore, the final decimal digit of the fourth power of an integer can only be 0, 1, 5, or 6. ■

Question 14

Prove that there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$.

For the sake of contradiction, assume that there exist integers x and y such that $2x^2 + 5y^2 = 14$.

Since integers are closed under multiplication, x^2 and y^2 are integers.

Since $2x^2$ is an integer multiple of 2, it must be even.

Since $2x^2$ is even, $5y^2$ must also be even for the sum of the two terms to be even.

This implies that both x and y must be even.

Let $x = 2m$ and $y = 2n$, where m and n are integers.

Substitute $x = 2m$ and $y = 2n$ into the equation $2x^2 + 5y^2 = 14$:

$$2(2m)^2 + 5(2n)^2 = 8m^2 + 20n^2 = 4(2m^2 + 5n^2) = 14$$

This implies that $2m^2 + 5n^2 = \frac{14}{4} = \frac{7}{2}$, which is not an integer.

Therefore, there are no solutions in integers x and y to the equation $2x^2 + 5y^2 = 14$. ■

Question 15

Prove that there are infinitely many solutions in positive integers x , y , and z to the equation $x^2 + y^2 = z^2$. [Hint: Let $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, where m and n are integers.]

Let $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$, where m and n are integers.

Substitute these values into the equation $x^2 + y^2 = z^2$:

$$(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2$$

Expanding the left side:

$$(m^4 - 2m^2n^2 + n^4) + 4m^2n^2 = m^4 + 2m^2n^2 + n^4$$

Simplifying:

$$m^4 - 2m^2n^2 + n^4 + 4m^2n^2 = m^4 + 2m^2n^2 + n^4$$

$$m^4 + 2m^2n^2 + n^4 = m^4 + 2m^2n^2 + n^4$$

Therefore, the equation $x^2 + y^2 = z^2$ holds true for $x = m^2 - n^2$, $y = 2mn$, and $z = m^2 + n^2$.

Since m and n can take on infinitely many integer values, there are infinitely many solutions in positive integers x , y , and z to the equation $x^2 + y^2 = z^2$. ■